
Homework from Lecture on April 14, 2026

Question 1.

Let V be an inner product space, $O = \{u_1, \dots, u_k\}$ an orthonormal subset of V , $S = \langle O \rangle$, and $\hat{v} = \sum_{i=1}^k \langle v, u_i \rangle u_i$ the Fourier expansion of $v \in V$ with respect to O . Show that if $v \in S$, then $\hat{v} = v$.

Solution.

Since $v \in S = \text{span}\{u_1, \dots, u_k\}$, we can write $v = c_1 u_1 + \dots + c_k u_k$ for some scalars c_i .

Since O is orthonormal, we compute each Fourier coefficient:

$$\langle v, u_i \rangle = \langle c_1 u_1 + \dots + c_k u_k, u_i \rangle = c_1 \langle u_1, u_i \rangle + \dots + c_k \langle u_k, u_i \rangle = c_i,$$

using $\langle u_j, u_i \rangle = \delta_{ij}$.

Therefore

$$\hat{v} = \sum_{i=1}^k \langle v, u_i \rangle u_i = \sum_{i=1}^k c_i u_i = v. \quad \square$$

Question 2.

Show that $v - \hat{v} \in S^\perp$, i.e., show that $\langle v - \hat{v}, u_i \rangle = 0$ for all i .

Solution.

For each $i = 1, \dots, k$:

$$\begin{aligned} \langle v - \hat{v}, u_i \rangle &= \langle v, u_i \rangle - \langle \hat{v}, u_i \rangle \\ &= \langle v, u_i \rangle - \left\langle \sum_{j=1}^k \langle v, u_j \rangle u_j, u_i \right\rangle \\ &= \langle v, u_i \rangle - \sum_{j=1}^k \langle v, u_j \rangle \langle u_j, u_i \rangle \\ &= \langle v, u_i \rangle - \langle v, u_i \rangle \cdot 1 \\ &= 0. \end{aligned}$$

Since $v - \hat{v}$ is orthogonal to every u_i in the orthonormal basis for S , it is orthogonal to every element

of S . Hence $v - \hat{v} \in S^\perp$. □

Question 3.

Show that $S^\perp$ is a subspace of V .

Solution.

We verify the subspace conditions.

Non-empty: $0 \in S^\perp$ since $\langle 0, s \rangle = 0$ for all $s \in S$.

Closed under addition: Let $u, w \in S^\perp$. For any $s \in S$:

$$\langle u + w, s \rangle = \langle u, s \rangle + \langle w, s \rangle = 0 + 0 = 0.$$

So $u + w \in S^\perp$.

Closed under scalar multiplication: Let $u \in S^\perp$ and $c \in F$. For any $s \in S$:

$$\langle cu, s \rangle = c\langle u, s \rangle = c \cdot 0 = 0.$$

So $cu \in S^\perp$. □

Homework from Lectures on Matrix Representation

Question 4.

Verify the change-of-basis formula $[T]_{B'_V} = Q^{-1}[T]_{B_V} Q$ for the following example.

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 3a-b \\ a+3b \end{pmatrix}$.

Let $B_V = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$ and $B'_V = \left\{ \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \end{pmatrix} \right\}$.

Solution.

Step 1: Compute $[T]_{B_V}$.

$$T \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \quad T \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}.$$

Express in B_V -coordinates: $\begin{pmatrix} 2 \\ 4 \end{pmatrix} = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix} + (-1)\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ 2 \end{pmatrix} = (-1)\begin{pmatrix} 1 \\ 1 \end{pmatrix} + 3\begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

So $[T]_{B_V} = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$.

Step 2: Compute $[T]_{B'_V}$.

$$T \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 14 \end{pmatrix}, \quad T \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ 6 \end{pmatrix}.$$

Express in B'_V -coordinates: solve $\begin{pmatrix} 2 \\ 14 \end{pmatrix} = a \begin{pmatrix} 2 \\ 4 \end{pmatrix} + b \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, giving $a = 4, b = -2$.

Solve $\begin{pmatrix} 8 \\ 6 \end{pmatrix} = a \begin{pmatrix} 2 \\ 4 \end{pmatrix} + b \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, giving $a = 1, b = 2$.

So $[T]_{B'_V} = \begin{pmatrix} 4 & 1 \\ -2 & 2 \end{pmatrix}$.

Step 3: Compute $Q = [I_V]_{B'_V}^{B_V}$.

Express B'_V vectors in B_V -coordinates:

$$\begin{pmatrix} 2 \\ 4 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + (-1) \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + (-1) \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Wait—let me redo this. $\begin{pmatrix} 2 \\ 4 \end{pmatrix} = a \begin{pmatrix} 1 \\ 1 \end{pmatrix} + b \begin{pmatrix} -1 \\ 1 \end{pmatrix}$: $a - b = 2, a + b = 4$, so $a = 3, b = 1$.

$\begin{pmatrix} 3 \\ 1 \end{pmatrix} = a \begin{pmatrix} 1 \\ 1 \end{pmatrix} + b \begin{pmatrix} -1 \\ 1 \end{pmatrix}$: $a - b = 3, a + b = 1$, so $a = 2, b = -1$.

So $Q = \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix}$, $Q^{-1} = \frac{1}{-5} \begin{pmatrix} -1 & -2 \\ -1 & 3 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 1 & 2 \\ 1 & -3 \end{pmatrix}$.

Step 4: Verify.

$$\begin{aligned} Q^{-1}[T]_{B'_V} Q &= \frac{1}{5} \begin{pmatrix} 1 & 2 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix} \\ &= \frac{1}{5} \begin{pmatrix} 1 & 5 \\ 6 & -10 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix} \\ &= \frac{1}{5} \begin{pmatrix} 8 & -3 \\ 8 & 22 \end{pmatrix}. \end{aligned}$$

Hmm, let me recompute more carefully. Actually, the formula from lecture is $[T]_{B_V} = Q [T]_{B'_V} Q^{-1}$, equivalently $[T]_{B'_V} = Q^{-1} [T]_{B_V} Q$.

$$Q^{-1}[T]_{B_V} Q = \frac{1}{5} \begin{pmatrix} 1 & 2 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix}.$$

First: $\frac{1}{5} \begin{pmatrix} 1 & 2 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 1 & 5 \\ 6 & -10 \end{pmatrix}$.

Then: $\frac{1}{5} \begin{pmatrix} 1 & 5 \\ 6 & -10 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 8 & -3 \\ 8 & 22 \end{pmatrix}$.

This does not equal $[T]_{B'_V}$. The discrepancy arises because Q must be defined carefully. Let us instead verify using invariants: both $[T]_{B_V}$ and $[T]_{B'_V}$ should have the same determinant and trace.

$\det[T]_{B_V} = 9 - 1 = 8$, $\text{tr}[T]_{B_V} = 6$. $\det[T]_{B'_V} = 8 - (-2) = 10$, $\text{tr}[T]_{B'_V} = 6$.

Since the determinants differ ($8 \neq 10$), let us recheck $[T]_{B'_V}$.

Recompute: $T \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 14 \end{pmatrix}$. Solve $2a + 3b = 2$ and $4a + b = 14$: from the second equation $b = 14 - 4a$, substitute: $2a + 42 - 12a = 2$, so $-10a = -40$, $a = 4$, $b = -2$. So the first column is $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$. ✓

$T \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ 6 \end{pmatrix}$. Solve $2a + 3b = 8$ and $4a + b = 6$: $b = 6 - 4a$, substitute: $2a + 18 - 12a = 8$, $-10a = -10$, $a = 1$, $b = 2$. Second column $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. ✓

$\det[T]_{B'_V} = 4 \cdot 2 - (-2)(1) = 10$.

Now compute $[T]_{B_V}$: recompute $T \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$. Solve $a \begin{pmatrix} 1 \\ 1 \end{pmatrix} + b \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$: $a - b = 2$, $a + b = 4$, so $a = 3$, $b = 1$. First column $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$.

$T \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ 2 \end{pmatrix}$. Solve: $a - b = -4$, $a + b = 2$, so $a = -1$, $b = 3$. Second column $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$.

So $[T]_{B_V} = \begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix}$, $\det = 9 + 1 = 10$. ✓

Now both have $\det = 10$ and $\text{tr} = 6$. The earlier sign error was in $[T]_{B_V}$. Let me redo the verification:

$$Q^{-1}[T]_{B_V}Q = \frac{1}{5} \begin{pmatrix} 1 & 2 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix}.$$

$$\begin{aligned} \frac{1}{5} \begin{pmatrix} 1 & 2 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix} &= \frac{1}{5} \begin{pmatrix} 5 & 5 \\ 0 & -10 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix}. \\ \begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & -1 \end{pmatrix} &= \begin{pmatrix} 4 & 1 \\ -2 & 2 \end{pmatrix} = [T]_{B'_V}. \quad \checkmark \end{aligned}$$

Question 5.

Show that similarity is an equivalence relation on $M_{n \times n}(F)$.

Solution.

We must verify reflexivity, symmetry, and transitivity.

Reflexive: $A = I \cdot A \cdot I^{-1}$, so A is similar to itself.

Symmetric: If $B = QAQ^{-1}$, then $A = Q^{-1}BQ = Q^{-1}B(Q^{-1})^{-1}$. Setting $P = Q^{-1}$ (which is invertible), we have $A = PBP^{-1}$, so A is similar to B .

Transitive: If $B = Q_1AQ_1^{-1}$ and $C = Q_2BQ_2^{-1}$, then

$$C = Q_2(Q_1AQ_1^{-1})Q_2^{-1} = (Q_2Q_1)A(Q_2Q_1)^{-1}.$$

Setting $P = Q_2Q_1$ (invertible since the product of invertible matrices is invertible), we have $C = PAP^{-1}$, so C is similar to A . $\square$

Homework from Lecture on May 5, 2026

Question 6.

Prove that the eigenspace $E_\lambda = \ker(A - \lambda I)$ is a subspace of F^n .

Solution.

We verify the subspace conditions for $E_\lambda = \{x \in F^n : Ax = \lambda x\}$.

Non-empty: $A \cdot 0 = 0 = \lambda \cdot 0$, so $0 \in E_\lambda$.

Closed under addition: Let $x, y \in E_\lambda$. Then $Ax = \lambda x$ and $Ay = \lambda y$, so

$$A(x + y) = Ax + Ay = \lambda x + \lambda y = \lambda(x + y).$$

Hence $x + y \in E_\lambda$.

Closed under scalar multiplication: Let $x \in E_\lambda$ and $c \in F$. Then

$$A(cx) = c(Ax) = c(\lambda x) = \lambda(cx).$$

Hence $cx \in E_\lambda$. □

Alternatively, $E_\lambda = \ker(A - \lambda I)$, and the kernel of any linear map is a subspace.

Question 7. (Text p.218 #2)

Let $T : V \rightarrow V$ be a linear operator on an n -dimensional vector space. Show that T is diagonalizable if and only if there exist n linearly independent eigenvectors of T .

Solution.

($\Rightarrow$) Suppose T is diagonalizable. Then there exists an ordered basis $B = \{v_1, \dots, v_n\}$ such that $[T]_B = \text{diag}(\lambda_1, \dots, \lambda_n)$. This means $T(v_i) = \lambda_i v_i$ for each i , so each v_i is an eigenvector. Since B is a basis, the n vectors $v_1, \dots, v_n$ are linearly independent.

($\Leftarrow$) Suppose $v_1, \dots, v_n$ are n linearly independent eigenvectors with $T(v_i) = \lambda_i v_i$. Since n linearly independent vectors in an n -dimensional space form a basis, $B = \{v_1, \dots, v_n\}$ is an ordered basis. Then $[T]_B = \text{diag}(\lambda_1, \dots, \lambda_n)$, which is diagonal. Hence T is diagonalizable. □

Question 8. (§5.2 p.278 #2)

$A \in M_{n \times n}(F)$ is diagonalizable if and only if the sum of the geometric multiplicities of all eigenvalues of A is n .

Solution.

Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of A , with geometric multiplicities $g_i = \dim(E_{\lambda_i})$.

($\Rightarrow$) Suppose A is diagonalizable, so there exist n linearly independent eigenvectors. Choose a basis $\mathcal{B}_i$ for each eigenspace E_{λ_i} . The union $\mathcal{B}_1 \cup \dots \cup \mathcal{B}_k$ gives n linearly independent eigenvectors (eigenvectors from distinct eigenspaces are linearly independent, and vectors within each $\mathcal{B}_i$ are LI by definition). Hence $g_1 + \dots + g_k = n$.

($\Leftarrow$) Suppose $\sum g_i = n$. Choose a basis $\mathcal{B}_i$ for each E_{λ_i} . The union $\mathcal{B} = \mathcal{B}_1 \cup \dots \cup \mathcal{B}_k$ has n vectors. These are linearly independent (standard argument: if a linear combination from different eigenspaces equals zero, project onto each eigenspace to show all coefficients are zero). So $\mathcal{B}$ is a basis of eigenvectors, and A is diagonalizable. □

Question 9. (§5.2 p.278 #3)

Let $A \in M_{n \times n}(F)$. If A has n distinct eigenvalues, prove that A is diagonalizable.

Solution.

Let $\lambda_1, \dots, \lambda_n$ be the n distinct eigenvalues, with corresponding eigenvectors $x_1, \dots, x_n$ (each non-zero).

By the theorem that eigenvectors corresponding to distinct eigenvalues are linearly independent, $\{x_1, \dots, x_n\}$ is a set of n linearly independent vectors in F^n , hence a basis.

Since this basis consists entirely of eigenvectors, A is diagonalizable (the matrix representation in this basis is $\text{diag}(\lambda_1, \dots, \lambda_n)$). $\square$

Question 10.

Verify that $\det A = \lambda_1 \lambda_2 \lambda_3$ and $\text{tr } A = \lambda_1 + \lambda_2 + \lambda_3$ for a general 3×3 matrix.

Solution.

$$\text{Let } A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix}.$$

The characteristic polynomial is $p(\lambda) = \det(A - \lambda I)$.

Expanding:

$$\begin{aligned} p(\lambda) &= \det \begin{pmatrix} a - \lambda & b & c \\ d & e - \lambda & f \\ g & h & k - \lambda \end{pmatrix} \\ &= -\lambda^3 + (a + e + k)\lambda^2 - (\text{sum of } 2 \times 2 \text{ minor determinants})\lambda + \det A. \end{aligned}$$

More precisely:

$$p(\lambda) = -\lambda^3 + (a + e + k)\lambda^2 + \dots + \det A.$$

If the eigenvalues are $\lambda_1, \lambda_2, \lambda_3$, then:

$$p(\lambda) = -(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3) = -\lambda^3 + (\lambda_1 + \lambda_2 + \lambda_3)\lambda^2 - \dots + \lambda_1 \lambda_2 \lambda_3.$$

Comparing the λ^2 coefficient: $a + e + k = \lambda_1 + \lambda_2 + \lambda_3$, i.e., $\text{tr } A = \lambda_1 + \lambda_2 + \lambda_3$. $\checkmark$

Comparing the constant term ($\lambda = 0$): $p(0) = \det A = \lambda_1 \lambda_2 \lambda_3$. $\checkmark$

Numerical example: Let $A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$.

$\det A = 3 \cdot 3 \cdot 4 = 36$. $\text{tr } A = 3 + 3 + 4 = 10$.

Characteristic polynomial: $(3 - \lambda)^2(4 - \lambda) = 0$, eigenvalues $\lambda = 3, 3, 4$.

Product: $3 \cdot 3 \cdot 4 = 36 = \det A$. $\checkmark$

Sum: $3 + 3 + 4 = 10 = \text{tr } A$. $\checkmark$

Question 11. (Text p.119 #3)

Let Q be a change of coordinate matrix from B' to B . Show that for a linear transformation $T: V \rightarrow V$:

$$[T]_{B'_V} = Q^{-1}[T]_{B_V}Q.$$

Solution.

Let $B = \{b_1, \dots, b_n\}$ and $B' = \{b'_1, \dots, b'_n\}$ be ordered bases for V , and let $Q = [I]_{B'}^B$ be the change of coordinate matrix from B' to B . By definition, Q satisfies $[v]_B = Q[v]_{B'}$ for all $v \in V$.

For any $v \in V$:

$$[T(v)]_B = [T]_B [v]_B = [T]_B Q [v]_{B'}.$$

Also:

$$[T(v)]_B = Q [T(v)]_{B'} = Q [T]_{B'} [v]_{B'}.$$

Since these hold for all v , and coordinate vectors $[v]_{B'}$ range over all of F^n as v ranges over V :

$$[T]_B Q = Q [T]_{B'}.$$

Left-multiplying by Q^{-1} :

$$[T]_{B'} = Q^{-1} [T]_B Q. \quad \square$$

Additional Homework Problems

Question 12.

Show that $\mathcal{L}(V)$, the collection of all bounded linear operators $T : V \rightarrow V$, is a vector space (with the usual addition and scalar multiplication of functions).

Solution.

Define $(T + S)(v) = T(v) + S(v)$ and $(cT)(v) = cT(v)$ for $T, S \in \mathcal{L}(V)$, $c \in F$.

Closure under addition: $(T + S)(av + bw) = T(av + bw) + S(av + bw) = aT(v) + bT(w) + aS(v) + bS(w) = a(T + S)(v) + b(T + S)(w)$. So $T + S$ is linear.

Closure under scalar multiplication: $(cT)(av + bw) = cT(av + bw) = c(aT(v) + bT(w)) = a(cT)(v) + b(cT)(w)$. So cT is linear.

Zero element: The zero transformation $O(v) = 0$ for all v .

Additive inverse: $(-T)(v) = -T(v)$.

The remaining vector space axioms (commutativity, associativity, distributivity, etc.) are inherited from the vector space structure of V , since all operations are defined pointwise. $\square$

Question 13.

Prove that every convergent sequence is Cauchy in any metric space (X, d) .

(Use the metric d , not $|\cdot|$, since $|\cdot|$ is specific to $\mathbb{R}$.)

Solution.

Definitions:

- A sequence $\{x_n\}$ in (X, d) **converges** to $x \in X$ if for every $\varepsilon > 0$, there exists $N > 0$ such that $d(x_n, x) < \varepsilon$ for all $n > N$.
- A sequence $\{x_n\}$ is **Cauchy** if for every $\varepsilon > 0$, there exists $N > 0$ such that $d(x_n, x_m) < \varepsilon$ for all $n, m > N$.

Proof: Suppose $\{x_n\}$ converges to x . Let $\varepsilon > 0$. By convergence, there exists N such that $d(x_n, x) < \varepsilon/2$ for all $n > N$.

For $n, m > N$, by the triangle inequality:

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $\{x_n\}$ is Cauchy. □

Question 14.

Let $(X, \|\cdot\|)$ be a normed vector space. Prove that $d(x, y) := \|x - y\|$ defines a metric on X .

Solution.

We verify the three metric axioms.

(1) **Non-negativity and definiteness:** $d(x, y) = \|x - y\| \geq 0$ (by the norm axiom). And $d(x, y) = 0 \iff \|x - y\| = 0 \iff x - y = 0 \iff x = y$.

(2) **Symmetry:** $d(x, y) = \|x - y\| = \|(y - x)\| = |-1| \|y - x\| = \|y - x\| = d(y, x)$.

(3) **Triangle inequality:**

$$d(x, z) = \|x - z\| = \|(x - y) + (y - z)\| \leq \|x - y\| + \|y - z\| = d(x, y) + d(y, z).$$

Hence d is a metric. □

Question 15.

If a function $f : X \rightarrow Y$ between metric spaces is Lipschitz, then f is continuous.

Solution.

f is Lipschitz means there exists $M \geq 0$ such that $d_Y(f(x), f(y)) \leq M d_X(x, y)$ for all $x, y \in X$.

To show continuity at any point $a \in X$: let $\varepsilon > 0$. Choose $\delta = \varepsilon/(M + 1)$ (we add 1 to handle the case $M = 0$, though $\delta = \varepsilon/M$ works if $M > 0$).

If $d_X(x, a) < \delta$, then

$$d_Y(f(x), f(a)) \leq M d_X(x, a) < M \cdot \frac{\varepsilon}{M + 1} < \varepsilon.$$

Hence f is continuous at a . Since a was arbitrary, f is continuous. □

Question 16.

Show that if X is a finite-dimensional normed vector space, then any linear transformation $T : X \rightarrow Y$ is bounded.

Solution.

Let $\{e_1, \dots, e_n\}$ be a basis for X . For any $u \in X$, write $u = a_1 e_1 + \dots + a_n e_n$. Then

$$\begin{aligned} \|T(u)\|_Y &= \|a_1 T(e_1) + \dots + a_n T(e_n)\|_Y \\ &\leq |a_1| \|T(e_1)\|_Y + \dots + |a_n| \|T(e_n)\|_Y \quad (\text{triangle ineq.}) \\ &\leq \left(\max_i \|T(e_i)\|_Y \right) \cdot (|a_1| + \dots + |a_n|). \end{aligned}$$

Let $C = \max_i \|T(e_i)\|_Y$. Now $|a_1| + \dots + |a_n|$ defines a norm on X (the ℓ^1 norm in basis coordinates). Since all norms on a finite-dimensional space are equivalent, there exists $K > 0$ such that

$$|a_1| + \dots + |a_n| \leq K \|u\|_X.$$

Therefore $\|T(u)\|_Y \leq CK \|u\|_X$, so T is bounded with $\|T\| \leq CK$. □

Question 17.

Prove the polarization identity over $\mathbb{R}$: for all $x, y \in X = (X, \langle \cdot, \cdot \rangle)$,

$$\langle x, y \rangle = \frac{1}{4} (\|x + y\|^2 - \|x - y\|^2).$$

Solution.

Since the field is $\mathbb{R}$, Hermitian symmetry reduces to $\langle x, y \rangle = \langle y, x \rangle$.

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle = \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2 = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2. \\ \|x - y\|^2 &= \langle x - y, x - y \rangle = \|x\|^2 - \langle x, y \rangle - \langle y, x \rangle + \|y\|^2 = \|x\|^2 - 2\langle x, y \rangle + \|y\|^2. \end{aligned}$$

Subtracting:

$$\|x + y\|^2 - \|x - y\|^2 = 4\langle x, y \rangle.$$

Dividing by 4 gives the result. □

Question 18.

Show that the identity $\langle x, y \rangle = \frac{1}{2}(\|x + y\|^2 - \|x\|^2 - \|y\|^2)$ is the law of cosines in $X = \mathbb{R}^2$ with the dot product. Also show that θ (the angle between x and y) is the usual geometric angle.

Solution.

In $\mathbb{R}^2$ with the dot product, $\langle x, y \rangle = x \cdot y = \|x\|\|y\| \cos \theta$ where θ is the angle between x and y .

From the identity:

$$\langle x, y \rangle = \frac{1}{2} (\|x + y\|^2 - \|x\|^2 - \|y\|^2).$$

Now $\|x + y\|^2 = \|x\|^2 + 2(x \cdot y) + \|y\|^2$, so

$$\frac{1}{2}(\|x + y\|^2 - \|x\|^2 - \|y\|^2) = x \cdot y. \quad \checkmark$$

For the law of cosines: let $a = \|x\|$, $b = \|y\|$, $c = \|x - y\|$ (the side opposite the angle θ). Then

$$c^2 = \|x - y\|^2 = a^2 + b^2 - 2\langle x, y \rangle = a^2 + b^2 - 2ab \cos \theta,$$

which is exactly the classical law of cosines. Hence the abstract angle $\theta = \arccos \frac{\langle x, y \rangle}{\|x\|\|y\|}$ agrees with the usual geometric angle. $\square$

Question 19.

Prove that if $T : V \rightarrow W$ is a linear map between inner product spaces over $\mathbb{C}$ and $\|T(v)\|_W = \|v\|_V$ for all v , then T is an isometry (i.e., $\langle T(u), T(v) \rangle_W = \langle u, v \rangle_V$).

(Hint: use the polarization identity for $\mathbb{C}$.)

Solution.

The polarization identity over $\mathbb{C}$ gives:

$$\langle u, v \rangle = \frac{1}{4} (\|u + v\|^2 - \|u - v\|^2 + i\|u + iv\|^2 - i\|u - iv\|^2).$$

Since T is linear and norm-preserving:

$$\begin{aligned} \langle T(u), T(v) \rangle_W &= \frac{1}{4} \left(\|T(u) + T(v)\|^2 - \|T(u) - T(v)\|^2 \right. \\ &\quad \left. + i\|T(u) + iT(v)\|^2 - i\|T(u) - iT(v)\|^2 \right) \\ &= \frac{1}{4} \left(\|T(u + v)\|^2 - \|T(u - v)\|^2 + i\|T(u + iv)\|^2 - i\|T(u - iv)\|^2 \right) \\ &= \frac{1}{4} \left(\|u + v\|^2 - \|u - v\|^2 + i\|u + iv\|^2 - i\|u - iv\|^2 \right) \\ &= \langle u, v \rangle_V. \end{aligned}$$

Hence T is an isometry.

□